

Popular OSS Tools for DevOps Realisation

DevOps is not just a buzz word today but is a methodology adopted increasingly by the IT industry for software development and infrastructure management. This article talks about the various aspects of DevOps realisation and the open source DevOps tools that are popular in the industry today.



Enterprises today are increasingly making the DevOps paradigm shift and one does not see a huge gap between the development and operation functions like before. When it comes to software development and infrastructure management methodologies, the old ways have been replaced with agile practices. In this article, I intend to make readers appreciate the various aspects of DevOps realisation and the tools in the open source world that are most popular in the DevOps adoption strategy.

Introducing DevOps as a cultural philosophy

DevOps is not merely a methodology of practices and tools. It is a step beyond that – it is a cultural philosophy that greatly enhances an organisation's ability to deliver applications and services with increased agility and high quality, by bringing in a mindset

of collaboration between traditionally siloed teams. The development and operations teams in the DevOps culture no longer have separate roles and skills that define them. Products are delivered by seamless integration of technologies that require greater insights into both the development and operational aspects of the product. The teams share responsibilities for what they build, improving transparency, communication and collaboration, thereby contributing to the quality and speed of delivery of the final product.

Infrastructure as a Code and OSS tools

Infrastructure as a Code or IaC is a terminology used in DevOps to describe the model used for the management of infrastructure (networks, load balancers, virtual machines, connections, etc) using the same versioning method that

applications and code use for general release. This helps deploy software faster in multiple servers avoiding inconsistencies of software versions in the servers, thereby improving productivity and reducing costs.

Through IaC, servers and other infrastructure are managed by automation scripts. This makes it faster and easier to maintain servers, maintain consistencies of environments (Dev, QA, Staging, UAT, Prod), reduce risks and manual mistakes, improve overall quality, reduce costs and save time owing to automation of repeated tasks.

There are many IaC tools that come under the umbrella of configuration management. These tools help configure the software systems on servers that are already provisioned. For example, automating installation of packages, starting services and placing configuration files on the system come under configuration management.

